DHRUV PANWAR

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EDUCATION

Vellore Institute of Technology (VIT) 2023 – 2027
B.Tech, Computer Science Engineering (E-Commerce Technology) — CGPA: 8.16/10

SS Children's Academy, Moradabad, U.P. (CBSE) 2022 – 2023
Class XII (PCM) — 83.8%

Podar International School, Ahmedabad (CBSE) 2021 – 2022
Class X — 88.4%

TECHNICAL SKILLS

- **Languages:** C++ (Programming), Python (Project Development), SQL
- **CS Fundamentals:** Data Structures & Algorithms, Object-Oriented Design, Operating Systems, DBMS, Complexity Analysis
- **Competitive Programming:** 400+ problems solved across LeetCode, Codeforces, and CodeChef
- **Tools & Platforms:** Git, GitHub, Docker, FastAPI, CLI, Streamlit, Jupyter
- **AI/ML (Applied):** FastEmbed (ONNX), LangChain, FAISS, YOLOv8, OpenCV, scikit-learn, NumPy, Pandas, Google Gemini API

PROJECTS

VisionMind AI: Intelligent Image Analysis Service GitHub
Live Demo: image-recognition-model-9l2e.onrender.com | *Tools:* Python, FastAPI, YOLOv8, Docker, Gemini, OpenCV, ReportLab

- Built and deployed a containerized computer vision microservice using FastAPI and Docker, exposing 12 REST endpoints with interactive Swagger documentation and sub-300ms CPU inference across 80 object classes at 640×640 resolution.
- Reduced cloud latency from ~35s to ~1.2s on a shared 0.5 vCPU host using browser-side canvas downscaling (97% payload reduction, ~4MB to ~120KB) and bundling in-memory annotations into a single request.
- Integrated Google Gemini for visual scene understanding with sequential fallback routing to handle HTTP 429 rate limits; generated spatial density heatmaps and streamed multi-page ReportLab PDF inspection briefs in $<$600ms.

DocuVault AI: Grounded Multi-Document RAG Microservice GitHub
Live Demo: docuvault-ai-ah83.onrender.com | *Tools:* Python, FastAPI, LangChain, FAISS, FastEmbed (ONNX), Docker, ReportLab

- Engineered a decoupled RAG microservice with 1,000-character semantic chunking (200-character overlap) and in-memory FAISS indexing, parsing multi-PDF uploads directly from RAM byte buffers without temporary disk I/O.
- Optimized embeddings using local ONNX CPU inference (384-dimensional all-MiniLM-L6-v2 via FastEmbed), reducing peak indexing RAM by 53% (~500MB to ~235MB on 512MB hosts) while delivering sub-150ms vector retrieval.
- Implemented sequential Gemini model fallback, L2 distance confidence tiers (High: $\leq$0.5, Medium: $\leq$1.0), an interactive chunk inspector, and automated PDF briefs; validated with 7 automated tests (100% CI pass rate).

EXPERIENCE

Software Engineering Intern – MPOnline Limited (TCS Joint Venture) Remote | May 2026 – Jul 2026

- Applied agile engineering practices across module refactoring and REST API workflows; maintained 15+ Git feature branches and validated core business logic with unit test suites achieving a 95%+ pass rate.

AI/ML Intern – MPOnline Limited (TCS Joint Venture) Remote | May 2026 – Jul 2026

- Engineered preprocessing, anomaly detection, and feature extraction pipelines across 10,000+ structured records; evaluated predictive baselines using Precision, Recall, and F1-score metrics.

CERTIFICATIONS & LANGUAGES

- **Certifications:** Oracle Cloud Infrastructure 2025 Certified Generative AI Professional (Oct 2025); NPTEL Marketing Management II (96%, Top 2 Percentile, National Elite Certificate)
- **Languages:** English (Full Professional), Hindi (Native), Gujarati (Limited Working)